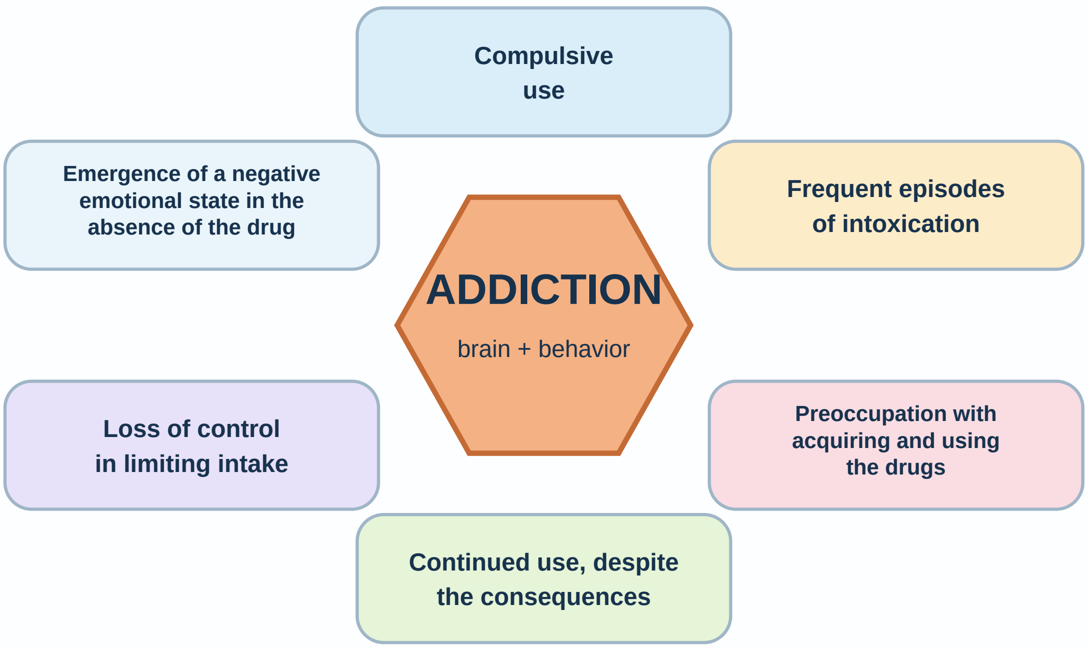
Understanding Addiction

Addiction is not just a “bad habit.”
It is a brain-and-behavior pattern where the reward system gets trained to keep returning to a substance or behavior, even when it causes harm.

Addiction at a glance



What usually happens

1. Relief or pleasure The substance gives a fast reward or escape.	**2. Brain learns it** The brain tags it as important and worth repeating.	**3. Craving grows** People, places, stress, or emotions can trigger urges.	**4. Control gets harder** The person may use again even when they do not want to.

Common signs of addiction

• Using more than planned or needing more to get the same effect.

• Thinking about the substance a lot or feeling strong cravings.

• Hiding use, lying, minimizing, or becoming defensive.

• Using despite problems with health, work, school, money, or relationships.

• Taking risks or doing things the person normally would not do.

What to remember

• Addiction often starts because something “works” at first: relief, numbness, energy, confidence, or escape.

• Over time, the brain begins to prioritize the substance over healthier rewards.

• Addiction is not something people are born with. Vulnerability to addiction develops through a combination of biological, environmental, and behavioral factors, including repeated choices and experiences that shape and train the brain over time.

• This is why people may keep using even when part of them wants to stop.

• Shame usually makes the cycle worse. Understanding helps treatment work better.

Helpful frame: **The problem is not that the person is weak.** The problem is that the brain has learned a very powerful shortcut to relief/reward.

How Addiction Works in the Brain

Dopamine / reward system
Addictive substances can create a bigger, faster, and more reliable dopamine surge than everyday rewards.

Nucleus accumbens
This is a key reward center. When it gets flooded with dopamine, the brain starts treating the substance like a high-priority need.

Hippocampus
Stores memories of the “good” or relieving feeling and remembers where, when, and with whom it happened.

Amygdala
Attaches emotion to those memories, so triggers can create craving later.

Prefrontal cortex
The part involved in planning, judgment, and stopping impulses can get overpowered, especially under stress.

Trigger stress, people, places, feelings	→	**Dopamine surge** fast reward in nucleus accumbens	→	**Memory + craving** hippocampus + amygdala	→	**Use again** even if there are consequences

Tolerance
Over time, the brain adapts. The same amount may feel weaker, so the person may use more to chase the same effect.

Why some things become highly addictive
The faster, stronger, and more predictable the dopamine release is, the more powerfully the brain learns it.

Recovery is possible
Brains can heal. Treatment helps people slow the cycle, manage triggers, build coping skills, and reconnect with healthy rewards.